GREG EZRA SEBASTIAN SITORUS

Data Scientist & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Science Technology undergraduate at Universitas Airlangga specializing in Machine Learning, Financial AI, Forensic Data Science, and End-to-End Data Engineering. Proven track record in building deterministic financial statement extraction engines (100% PSAK/IFRS accounting identity verification), graph-based & Benford's Law fraud detection pipelines, walk-forward quantitative backtesting platforms, and full-stack web applications (Python, Flask, R, SQL). National Finalist at the IFEST 2026 Data Analysis Competition with extensive hands-on experience across 7 national data science competitions (NLP, Computer Vision, Time-Series Forecasting, and Tabular Modeling).

EDUCATION

Universitas Airlangga (UNAIR) — *Faculty of Advanced Technology and Multidiscipline (FTMM), Surabaya* **Aug 2024 – Present**

- **Bachelor of Data Science Technology** (Student ID: 164241077) | GPA: 3.20 / 4.00
- **Relevant Coursework:** Data Mining I & II, Multivariate Analysis, Database Systems, Statistical Computing, Spatial Data Analysis, Exploratory Data Analysis & Visualization, Artificial Intelligence, Mathematical Statistics, Operations Research.

SMA Katolik Santo Carolus Surabaya — *High School Diploma, Surabaya* **2021 – 2024**

- **Continuous Academic Track at Santo Carolus Surabaya:** Senior High School (Graduated 2024), Junior High School (Graduated 2021), Elementary School (Graduated 2018) — completed all levels on schedule.

SCIENTIFIC RESEARCH & TECHNICAL WHITEPAPERS

Automated Financial Statement Intelligence for IDX (Reconciled-FinExtract) — *Financial NLP, Document AI, PSAK 1 / IFRS* **2026**

- **Deterministic Accounting Reconciliation:** Engineered an automated PDF financial statement extraction pipeline for Indonesia Stock Exchange (IDX) issuers integrated with a Double-Entry Accounting Reconciliation Engine enforcing PSAK 1 / IFRS identities (Assets = Liabilities + Equity; Cash Flow closure).
- **Zero Numeric Hallucination:** Built Indonesian locale numeric normalizers, accounting negative bracket parsers, and automated monetary scale detectors (thousands/millions/billions IDR), achieving 100% cell exact match and 0% numeric hallucination on audited IDX benchmarks (BBCA, ICBP, TLKM).

Algorithmic Detection of Bid Rigging in Public Procurement (LKPP / LPSE) — *Forensic Data Science & Graph Analytics* **2026**

- **Forensic Digit & Anomaly Auditing:** Developed an Early Warning System for collusive public procurement tenders combining Benford's Law first-digit tests (Chi-Square & Mean Absolute Deviation) to detect human-fabricated contract values and budget-splitting anomalies.
- **Bipartite Graph & Corruption Risk Index (CRI):** Modeled vendor-tender co-bidding networks, Shannon winner entropy, and composite red-flag risk scoring aligned with World Bank INT and OECD anti-cartel guidelines.

Tropical Deforestation & Wildfire Damage Assessment via Sentinel-2 & U-Net — *Geospatial AI & Remote Sensing* **2026**

- **High-Resolution Spatial Segmentation (10m):** Combined multi-temporal ESA Copernicus Sentinel-2 biophysical spectral indices (NDVI, NBR, dNBR) with a 6-channel PyTorch U-Net segmentation architecture, achieving 0.862 IoU and 0.925 Dice F1-Score for automated hectare-level burn scar quantification.

SOFTWARE, QUANTITATIVE & DATA ENGINEERING PROJECTS

StockLab & EquityLab — Quantitative Backtesting & Fundamental Screening Suite — *Python, Flask, Streamlit, Scikit-learn, Plotly* **2026**

- **StockLab (Walk-Forward ML Backtester):** Built an IDX equity machine learning backtesting platform featuring walk-forward validation, transaction cost modeling, and falsification checks against look-ahead bias. Re-architected the UI from Streamlit to Flask + JavaScript for unified single-server deployment.
- **EquityLab (Sector-Aware Fundamental Engine):** Developed a financial ratio and percentile screening library featuring automated Q4 discrete quarter derivation (FY - Q1..Q3), gap-safe QoQ/YoY growth computation, outlier flagging, and 33 automated pytest unit tests.

Three-Tier Forestry ETL Pipeline & Enterprise Dashboard — *Python, PostgreSQL (Supabase), Laravel, Tailwind CSS* **Oct 2026**

- **Enterprise Application Integration (EAI):** Architected a staging -> master -> replica ETL pipeline integrating 5 heterogeneous forestry datasets (BPS & KLHK SiPongi in XLS, XLSX, CSV, JSON) with MD5 checksum incremental replication, one-way delta synchronization, and a live Laravel analytics dashboard.

Stokku POS, Dapur Lestari & Unified Portfolio Server — *Python, Flask, SQLite, Node.js, WebSocket, HTML/CSS/JS* **2026**

- **Full-Stack Business Applications:** Built a role-based Point-of-Sale & inventory management system with automated stock deduction (Stokku), a catering order & CSV reporting portal (Dapur Lestari), a 4-player real-time WebSocket game, and an automated PII/secret sanitization pipeline (DS-BUILD).

DATA SCIENCE COMPETITIONS & ACHIEVEMENTS

National Finalist — Data Analysis Competition IFEST 2026 — *Universitas Padjadjaran (UNPAD)* **Sep 2026**

- **Preliminary Round (0.94835 Public Macro-F1):** Engineered 233 handcrafted NLP features from scratch (TF-IDF, LSA, BM25, Word2Vec/FastText, entity & negation statistics) and a 12-model rank-averaged ensemble with article-grouped cross-validation to detect news headline-body entailment under strict no-pretrained-weights rules.
- **Final Round (0.71259 CV Average Precision):** Reconstructed 79,756 missing values and 8,032 false zeros using hydrological water/area balance equations, predicting next-month watershed (HUC12) water stress via a blend of LightGBM, XGBoost, Quantile GBDT, and custom Transformer with TreeSHAP interpretability.

National Data Mining & Machine Learning Competition Track Record — *7 Documented Competition Case Studies* **2025 – 2026**

- **ACTION 2026 (Multi-Label NLP, 0.93360 Public Macro-F1) & ACTION 2025 Participant (UNESA):** Built a custom cipher/homoglyph/leetspeak normalizer and fine-tuned 5-fold group-split IndoBERT & IndoBERTweet ensembles with thread-context injection across 58,000 public service comments.
- **JOINTS x INSPIRE UGM 2026 (Time-Series Forecasting, 0.36493 Public MASE):** Developed a Distributional Hurdle LightGBM model with release-age augmentation (~500k training rows) and holiday decay adjustments (Eid/Ramadan/Nyepi) to forecast D4–D10 cinema ticket sales from D1–D3 history.
- **Computer Vision & Tabular Modeling (DATASCAPE, Satria Data, FIND IT, GAMMA FEST 2026):** Achieved 0.89063 Macro-F1 via manual/automated label-noise cleaning & ConvNeXt + CLIP ViT-L/14 blending (DATASCAPE); 0.9706 validation Macro-F1 with a 12-model 3-backbone x 4-fold ensemble (BDC Satria Data); 6-class Face Anti-Spoofing with Focal Loss & SWA (FIND IT); and evaluated 296 automated grid-search submissions using CatBoost Poisson & causal Elo form features (GAMMA FEST).

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Faculty Orientation Committee (PKKMB / PD FTMM) — Universitas Airlangga — *Event Division Staff (Staf Acara)* **2026**

- Designed event rundowns, managed stage flow and scheduling, and coordinated across divisions to execute faculty-wide orientation programs for incoming FTMM students.

Philanthropy — Community Service Department (Sigma), HIMATESDA UNAIR — *Organizing Committee (Logistics, Field Operations & Equipment Support)* **Nov 2025**

- Managed catering logistics while actively supporting Field Coordinators (Korlap) and Equipment (Perkap) teams in on-site crowd flow and technical setup for community service in Manukan Kulon, Surabaya (Certificate No: 5486/B/UN3.FTMM/KM.06.02/2025).

TECHNICAL SKILLS & CERTIFICATIONS

- **Programming & Databases:** Python (Advanced), R (Advanced), SQL (PostgreSQL, SQLite), JavaScript (ES6+), HTML5/CSS3, PHP (Laravel).
- **Machine Learning & Deep Learning:** Scikit-learn, LightGBM, XGBoost, CatBoost, PyTorch, timm, HuggingFace Transformers (IndoBERT, FinBERT, CLIP, ViT, U-Net), SHAP, Optuna.
- **Statistics, Analytics & Data Engineering:** Time-Series Forecasting, NLP, Computer Vision, Forensic Data Analysis (Benford's Law, Graph Theory), Multivariate Statistics (MANOVA, Hotelling T^2), Spatial Statistics (GWR/MGWR), ETL Pipelines, Pandas, NumPy, SciPy, Tidyverse, R Shiny, Streamlit, Plotly.
- **Certifications & Awards:** Finalist — Data Analysis Competition IFEST 2026 (FMIPA UNPAD); National Participant — Academic Competition of Data Science (ACTION) 2025 (FMIPA UNESA); Committee Certificate — Philanthropy FTMM UNAIR (2025).